SEMESTER - V

COURSE: MAJOR – 8 (MJ-8) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: ELECTROMAGNETIC THEORY **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to equip students with a deep understanding of electromagnetic theory, focusing on Maxwell's equations and their applications in various media. Through theoretical lectures, students will gain proficiency in analyzing electromagnetic wave propagation, polarization phenomena, and the behavior of electromagnetic waves in different materials and structures.

Learning Outcomes:

1. Understand the derivation of Maxwell's equations and the significance of displacement current in electromagnetism.
2. Analyze the concepts of vector and scalar potentials and apply boundary conditions at interfaces between different media.
3. Describe the characteristics of plane electromagnetic waves in unbounded and bounded media.
4. Analyze polarization phenomena, including linear, circular, and elliptical polarization, and their applications in optical devices.
5. Describe the propagation of electromagnetic waves in waveguides, including planar optical waveguides, and calculate phase and group velocities of guided waves.
6. Explain the principles of optical fibers, including numerical aperture, step and graded indices, and differentiate between single and multiple mode fibers.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Maxwell Equations (10 Lectures): Derivation of Maxwell's equations, Displacement Current, Vector and Scalar Potentials, Boundary Conditions at Interface between Different Media, Wave Equations, Plane Waves in Dielectric Media, Poynting Theorem and Poynting Vector, Electromagnetic (EM) Energy Density.

UNIT-II:

EM Wave Propagation in Unbounded and Bounded Media (20 Lectures): Plane EM waves through vacuum and isotropic dielectric medium, transverse nature of plane EM waves, refractive index and dielectric constant, wave impedance. Propagation through conducting media, relaxation time, skin depth. Boundary conditions at a plane interface between two media. Reflection & Refraction of plane waves at plane interface between two dielectric media-Laws of Reflection & Refraction. Fresnel's Formulae for perpendicular & parallel polarization cases, Brewster's law, Reflection & Transmission coefficients. Waveguides: Planar optical wave guides, Planar dielectric wave guide, Condition of continuity at interface, Phase shift on total reflection, Eigenvalue equations, Phase and group velocity of guided waves, Field energy and Power transmission. Numerical Aperture, Step and Graded Indices, Single and Multiple Mode Fibres.

UNIT-III:

Polarization of Electromagnetic Waves (15 Lectures): Description of Linear, Circular, and Elliptical Polarization, Uniaxial and Biaxial Crystals, Light Propagation in Uniaxial Crystal, Double Refraction. Polarization by Double Refraction. Nicol Prism (construction and working), Ordinary & extraordinary refractive indices. Production & detection of Plane, Circularly, and Elliptically Polarized Light, Phase Retardation Plates: Quarter-Wave and Half-Wave Plates, Babinet Compensator and its Uses, Analysis of Polarized Light. Optical Rotation, Biot's Laws for Rotatory Polarization, Fresnel's Theory of optical rotation, Calculation of angle of rotation, Experimental verification of Fresnel's theory, Specific rotation.

SUGGESTED READINGS:

1. Introduction to Electrodynamics, D. J. Griffiths, 3rd Ed., 1998, Benjamin Cummings.
2. Introduction to Electromagnetic Theory, T.L. Chow, 2006, Jones & Bartlett Learning
3. Fundamentals of Electromagnetics, M.A.W. Miah, 1982, Tata McGraw Hill
4. Electromagnetic field Theory, R. S. Kshetrimayun, 2012, Cengage Learning
5. Electromagnetic Field Theory for Engineers & Physicists, G. Lehner, 2010, Springer

6. Optical Electronics, Ajoy Ghatak and K. Tyagrajan, Cambridge University Press.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Verification of Gauss's Law for Electric and Magnetic Fields using PhEt simulations.
2. Verify displacement current concept using an oscilloscope and capacitor with varying frequency.
3. Measure intensity of reflected and transmitted beams for different polarization angles.
4. Observe changes in polarization using an analyzer.
5. Verification of Snell's Law and Total Internal Reflection with Semi-Circular Glass Block.
6. Identification of Polarization Type (Linear, Circular, Elliptical)

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - V

COURSE: MAJOR – 9 (MJ-9) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: NUMERICAL TECHNIQUES AND PROGRAMMING **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

To equip students with the fundamentals of numerical methods and programming techniques essential for solving scientific problems. The course aims to bridge the gap between analytical mathematics and computational implementation using programming languages like Fortran, C++, or Python.

Learning Outcomes:

1. After successful completion of this course, students will be able to:
2. Analyze the accuracy and stability of numerical algorithms.
3. Solve algebraic and transcendental equations numerically.
4. Implement interpolation and numerical integration techniques.
5. Develop and execute programs for solving scientific problems using modern programming languages.
6. Apply numerical techniques for solving ordinary differential equations.
7. Write modular and efficient code using subroutines and functions.

PART 'A'

UNIT-I:

Errors, Approximation, roots of Equations and Interpolation (15 Lectures): Significant digits, round-off and truncation errors, error propagation. Direct methods: Gaussian elimination, LU decomposition. Iterative methods: Gauss-Seidel, Jacobi iteration. Eigenvalue problems: Power method, Jacobi method.

Bisection method, Newton-Raphson method, Fixed-point iteration, convergence criteria. **Interpolation**: Newton's forward and backward formulae.

UNIT-II:

Numerical Differentiation, Integration, Solution of ODE and curve fitting (15 Lectures): Trapezoidal rule, Simpson's rule, Gaussian quadrature. Solution of Ordinary Differential Equations: Euler's method, Runge-Kutta methods 4th order. Least-Squares Curve Fitting, Polynomial and linear regression.

UNIT-III:

Programming Fundamentals (Python) (15 Lectures): Introduction to Programming: Flowcharts, Algorithms, Program Structure. Data Types and Arithmetic: Integer & Floating-point operations, Typecasting, Precision. Input and Output Statements: Console I/O, Formatted Output. Control Structures: Conditional Statements, Loops and Iteration. Arrays: 1D and 2D arrays, Initialization, Access, and Applications in Numerical Context. Functions and Subroutines: Definition, Parameter Passing, Return Values, Scope, Recursion Basics.

SUGGESTED READINGS:

1. Rajaraman, V. "Computer Oriented Numerical Methods", PHI Learning.
2. Chapra, S.C., and Canale, R.P. "Numerical Methods for Engineers", McGraw-Hill Education.
3. Jain, M.K., Iyengar, S.R.K., and Jain, R.K. "Numerical Methods for Scientific and Engineering Computation", New Age International.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Write a program to find roots of a nonlinear equation using the Bisection method.
2. Write a program to find roots using the Regula-Falsi method.
3. Write a program to find roots using the Newton-Raphson method.
4. Solve a system of linear equations using Gauss Elimination method.
5. Implement the Gauss-Seidel iterative method to solve a system of linear equations.
6. Compute eigenvalues of a matrix using the Power method.
7. Perform interpolation using Newton's forward difference formula.
8. Perform interpolation using Lagrange's interpolation formula.

9. Numerical differentiation using finite difference methods.

10. Numerical integration using the Trapezoidal rule and Simpson's 1/3 rule.

11. Fit a straight line to data using the least squares method.

12. Solve an ordinary differential equation using Euler's method.

13. Solve an ordinary differential equation using Runge-Kutta 4th order method.

14. Write a program to perform matrix multiplication and inverse.

15. Simulate a simple harmonic oscillator numerically using Python or C++.

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SEMESTER - V

COURSE: MAJOR – 10 (MJ-10) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: CLASSICAL MECHANICS **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to provide students with a comprehensive understanding of classical mechanics, including Newtonian mechanics, Lagrangian dynamics, Hamiltonian dynamics, variational principles, canonical transformations, and special theory of relativity. Through theoretical lectures, students will develop a strong foundation in classical mechanics principles and their applications to various physical systems.

Learning Outcomes:

1. Understand the principles of classical mechanics, including Newton's laws of motion, and their application to the motion of point particles in external electric and magnetic fields.
2. Apply Lagrangian dynamics principles, including generalized coordinates, D' Alembert's Principle, and Lagrange's equations, to analyze the motion of simple mechanical systems and particles in electromagnetic fields.
3. Apply Hamiltonian dynamics principles, including generalized momentum, conservation theorems, and Hamilton's equations, to analyze the motion of particles in central force fields and electromagnetic fields.
4. Understand variational principles, including the calculus of variations, Euler-Lagrange equations, and Hamiltonian principle, and apply them to analyze mechanical systems and determine the path of least action.
5. Apply canonical transformations principles, including Legendre transformations and Poisson brackets, to analyze the transformation of coordinates and momenta in phase space.

6. Understand the postulates of the special theory of relativity, Lorentz transformations, and their consequences such as length contraction, time dilation, and addition of velocities and its realization in relativistic phenomena.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Classical Mechanics of Point Particles (10 Lectures): Review of Newtonian Mechanics; Application to the motion of a charge particle in external electric and magnetic fields- motion in uniform electric field, magnetic field- gyroradius and gyrofrequency, motion in crossed electric and magnetic fields. Coordinate Systems, Degrees of freedom, generalized coordinates and velocities, Principle of Virtual work, D' Alembert's Principle.

UNIT-II:

Lagrangian and Hamiltonian Dynamics (20 Lectures): Hamilton's principle and Lagrange's equations, Euler- Lagrange equations, Examples of the Euler-Lagrange equations: one- dimensional Simple Harmonic Oscillation and falling body in uniform gravity, Applications to simple systems such as coupled oscillators, Charges particle moving in an Electromagnetic field. Hamiltonian Dynamics: Generalized momentum and cyclic coordinates, Conservation Theorems, Hamiltonian's equations in Different Coordinate Systems. Examples in Hamiltonian Dynamics: 1-D and 2-D Harmonic Oscillator, Motion of a particle in a central force field, Charged particle moving in an electromagnetic field, Compound Pendulum.

UNIT-III:

Variational Principles and Canonical Transformations (15 Lectures): The Calculus of Variations and Euler-Lagrange's Equations, Hamiltonian Principle from D'Alembert's Principle, Modified Hamiltonian Principle, Lagrange's Method of Undetermined Multipliers: Simple Pendulum, Rolling hoop on an inclined Plane. Principle of Least Action. Canonical transformations and Legendre Transformations, Generating Functions, Condition for Canonical Transformations, Poisson's Brackets, Lagrange Brackets, Invariance of Poisson Bracket.

SUGGESTED READINGS:

1. Classical Mechanics, H.Goldstein, C.P. Poole, J.L. Safko, 3rd Edn. 2002, Pearson Education.
2. Classical Mechanics, J. C. Uppadhyaya, Himalaya Publising House Pvt. Ltd.
3. Mechanics, L. D. Landau and E. M. Lifshitz, 1976, Pergamon.

4. Classical Electrodynamics, J.D. Jackson, 3rd Edn., 1998, Wiley.

5. The Classical Theory of Fields, L.D Landau, E. M. Lifshitz, 4th Edn., 2003, Elsevier.

6. Introduction to Electrodynamics, D.J. Griffiths, 2012, Pearson Education.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Determination of moment of inertia of (a) Flywheel (b) Rod (c) Thin slab.
2. Determination of moment of inertia using Torsional Pendulum.
3. Simulate and analyze the linear motion of a charged particle under a uniform electric field using Newton's second law.
4. Simulate parabolic motion; analyze displacement, velocity, and acceleration using PhET simulation.
5. Verify the conservation laws in planetary motion using real data.

   Hints: Experimental data is available from **NASA JPL HORIZON** in text (ASCII) table format.
6. Experimentally investigate the motion of a compound pendulum and derive the corresponding Hamiltonian.
7. Experimentally and theoretically investigate the normal modes of two coupled pendulum.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B. L. Flint and H. T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D. P. Khandelwal, 1985, Vani Pub.

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SEMESTER - V

COURSE: MAJOR – 11 (MJ-11) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: QUANTUM MECHANICS-I **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to introduce students to the principles and concepts of quantum mechanics, highlighting its departure from classical mechanics and its applications in understanding the behavior of microscopic particles and systems.

Learning Outcomes:

1. Understand the inadequacy of classical mechanics in explaining certain phenomena and recognize the need for quantum mechanics.
2. Explain key concepts such as Planck's theory of blackbody radiation, the photoelectric effect, Compton scattering, and the De Broglie wavelength.
3. Analyze experimental evidence supporting wave-particle duality, including the Davisson-Germer experiment, and understand the implications of the Heisenberg uncertainty principle.
4. Describe the postulates of quantum mechanics and properties of wave functions, including normalization, linearity, superposition, and probability interpretation.
5. Solve the Schrödinger equation, both time-dependent and time-independent, for various systems including bound states in arbitrary potentials, and apply the solutions to analyze wave packet dynamics and energy eigenvalues.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Introduction to Quantum Mechanics (20 Lectures): Inadequacy of classical mechanics. Planck's theory of blackbody radiation. Photo-electric effect and Compton scattering. De- Broglie wavelength and matter waves. Davisson-Germer experiment. Group and Phase velocities and relation between them. Wave-particle duality, Heisenberg uncertainty principle. Consequences of the Uncertainty Principle- The path of an object, Zero Point Energy, The size of an atom, Existence of the electron inside the Nucleus. Postulates of Quantum Mechanics. Properties of Wave Function. Interpretation of Wave Function Probability and probability current densities in three dimensions; Conditions for Physical Acceptability of Wave Functions. Wave Function of a Free Particle. Normalization. Linearity and Superposition Principles. Eigenfunctions and Eigenvalues. Hermitian Operators. Commuting and Non-Commuting Operators. Position, momentum and Energy operators; commutator of position and momentum operators; Expectation values of position and momentum.

UNIT-II:

Schrodinger equation (15 Lectures): Time dependent Schrodinger equation and dynamical evolution of a quantum state. Time independent Schrodinger equation**.** Hamiltonian, stationary states and energy eigenvalues; expansion of an arbitrary wavefunction as a linear combination of energy eigenfunctions; General solution of the time dependent Schrodinger equation in terms of linear combinations of stationary states; Application to spread of Gaussian wave-packet for a free particle in one dimension; wave packets, Fourier transforms and momentum space wavefunction.

UNIT-III:

Applications of Schrodinger equation (10 Lectures) - Continuity of wavefunction, boundary condition and emergence of discrete energy levels; application to one-dimensional problem- Free particle in a box with rigid wall, infinite and finite square well potential, One dimensional square well, Linear harmonic oscillator (1D and 3D), 3D Problems: the free particle, the box potential, Rigid rotator and Hydrogen atom (s-state) ground state.

SUGGESTED READINGS:

1. Introduction to Quantum Mechanics, D.J. Griffith, 2nd Ed. 2005, Pearson Education
2. Quantum Mechanics, Eugen Merzbacher, 2004, John Wiley and Sons, Inc.

3. A Text book of Quantum Mechanics, P.M. Mathews and K. Venkatesan, 2nd Ed., 2010, McGraw Hill
4. Quantum Mechanics, Leonard I. Schiff, 3rd Edn. 2010, Tata McGraw Hill.
5. Quantum Mechanics, G. Aruldhas, 2nd Edn. 2002, PHI Learning of India.
6. Quantum Mechanics, Robert Eisberg and Robert Resnick, 2nd Edn., 2002, Wiley.
7. Quantum Mechanics Concepts and Applications 2nd Edition Nouredine Zettili.
8. Quantum Mechanics: Theory and Applications by Ajoy Ghatak, S. Lokanathan.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Simulate and analyze the spectral distribution of blackbody radiation and verify Planck's radiation law.
2. Study the variation of photoelectric current with frequency and intensity of light and determine Planck's constant.
3. Demonstrate the formation of wave packets and distinguish between group and phase velocities for matter waves.
4. Study the emission spectra of Hydrogen, Neon and mercury vapors.
5. Simulate how a free particle's Gaussian wave packet evolves and spreads in time.

SUGGESTED READINGS:

1. https://phet.colorado.edu/
2. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
3. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, Kitab Mahal
4. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
5. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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